

NFL PLAYER SALARY PREDICTION

Team Project – Georgia Tech M.S. in Analytics Program

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I. ABSTRACT

Our project aimed to model and predict the fully guaranteed value of an offensive NFL player's post-rookie deal; second contract. NFL contracts are complex and can be structured in a variety of ways; they may include any combination of base salary and bonuses. The base salary is lost by the player if they are cut by their team, unless there is a specified guaranteed amount. The guaranteed amount signals security for the player and team's willingness to provide a substantial up-front investment in the player without a guarantee of equal performance. Because of its importance to both the team and the player, our project focuses on predicting solely the guaranteed amount for NFL player contracts. Using data acquired from NFLverse, we performed exploratory data analysis and utilized the supervised learning methods of linear regression, random forest, support vector machine, gradient boosting, and generalized boosting regression for modeling.

Our data set was derived by combining offensive player performance statistics, contract, and draft data. The combined data was cleaned, duplicates were removed, and the entire data set was converted from a long format into a wide one, which was done to combine multiple rows of season data into a cumulative player set.

Our findings show that the random forest model performed the best in terms of root mean squared error as well as mean absolute error. The predictor `total_fantasy_points_1` was found to be especially important in our modeling, as it was the most significant predictor in three of the five models. It is no surprise that a fantasy football predictor would come out on top as it encapsulates many of the other statistics within it and is position agnostic. We also discovered that statistics from a player's final season before signing their second contract held more importance than statistics from prior seasons.

II. INTRODUCTION

The National Football League (NFL) is America's most popular sports league, and continues to grow in popularity, profitability, and globally. As a result, players have been signed to bigger and continually record-setting contracts. Using official NFL data made possible by the R package NFLverse, we are seeking to understand the offensive player performance statistics that are the most impactful and influential in the overall worth of an NFL contract.

Our analysis will focus on the overall contract guaranteed value of an offensive player based on performance metrics in the seasons leading up to their second NFL contract (post-rookie deal). In particular, we were interested in the criticality of "contract year" performance and will approach this analysis through the lens of various offensive positions; quarterback (QB), fullback (FB), running back (RB), tight end (TE), and wide receiver (WR).

By applying various models to the data, including linear regression, random forest, SVM, gradient boosting, and GBM, in conjunction with cross-validation, we will attempt to derive the most accurate model for prediction. We aim to predict an offensive player's next contract, in total guaranteed dollars, while also identifying contributing performance factors in a contract-year and the influence on second contracts.

III. PROBLEM STATEMENT / DATA SOURCES

Our data is sourced from the NFLverse package for R with data originating from the National Football League (NFL), Next Gen Stats, broadcast networks (CBS, ESPN, FOX, NBC, etc.), NFL teams and rosters, analytics providers (Pro Football Focus, SportsRadar), and other fan-driven platforms (Pro Football Reference).

Website: <https://nflverse.nflverse.com>

NFLverse data sets/repositories (and example attributes) used in our analysis and prediction (complete data dictionaries are included in the supplemental files):

- Contracts (2011 – 2023)
 - Value, in millions of dollars (the total contract value)
 - Years – length of the contract
 - APY, in millions of dollars (average payout per year)
 - Position (player position such as quarterback, running back, wide receiver, etc.)
 - Guaranteed, in millions of dollars (guaranteed contract value) – response variable
- Draft Picks (2011 – 2023)
 - Round (round in the draft the player was selected)
 - Pick (overall pick number)
 - Draft Team (team that selected the player)
 - Season (draft year)
- Player Stats (2011 – 2023)
 - Total Completions (total pass completions)
 - Total Attempts (total pass attempts)
 - Total Passing Yards (total yards from passes)
 - Total Passing TDs (total touchdowns from passes)
 - Total Interceptions (total passing turnovers)
 - Total Sacks (total sacks)
 - Total Rushing Yards (total rushing yards)
 - Total Rushing TDs (total touchdowns from rushes)
 - Total Receptions (total pass catches)
 - Total Targets (total passes targeting player)
 - Total Receiving Yards (total yards from pass catches)
 - Total Receiving TDs (total touchdowns from catches)
 - Total Receiving Yards After Catch (total yards accumulated after a catch)

a. DATA PREPARATION

Our initial dataset came from combining the Contracts, Draft Picks, and Player Stats datasets from the NFLverse repository. After combining, the dataset consisted of 24,498 rows representing 2,123 unique

players over 13 seasons. There were also 97 features in the initial dataset. From there, we took the following steps to produce a dataset that would be ready for analysis:

- Only included players with draft year of 2011 or later. This was to accommodate the Collective Bargaining Agreement (CBA) change that went into effect in 2011 and impacted contract structure and value. This change provided us with a player set operating under a consistent contract structure while still leaving us with enough players for analysis. We also excluded players with a draft year of 2021 or later because they largely have not signed their second contract yet.
- Only included players who play offensive positions (QB, FB, RB, TE, WR). Offensive and defensive statistics are very different in football, so removing defensive players allowed us to narrow the scope of statistics used in the analysis.
- Only included season statistics before a player's second contract. Since our analysis was aimed at predicting the value of a player's second contract, we removed statistics from seasons that occurred after the second contract was signed. For example, Cam Newton's statistics for 2011 – 2014 were retained as his second contract was signed in 2015.
- Removed unnecessary features. This included duplicate features from combining datasets (for example the "position" feature was in multiple datasets), defensive statistics, and full career statistics and awards.
- Following the steps above, we still had several rows for each player, with each row representing a different season. For example, Cam Newton had rows with statistics for each of his 2011-2014 seasons. To deal with this, we converted the dataset from a long format into a wide format. This ensured that each player only had one row (except for cases when the player was traded), and the statistics for each season were now features. To differentiate between seasons, we added a numerical suffix to each statistical feature noting how many years away those statistics were from the second contract being signed. For example, `total_rushing_yards_2` indicates the player's rushing yards in the season 2 years before their second contract was signed.
- Split the data into training (80%) and testing (20%) sets.
- Normalized the data so that all values were between 0 and 1. This step was only done for use in linear regression model.

Following all data preparation, our dataset consisted of 646 rows representing 580 unique players. As noted above, players who were traded before signing their second contract had a row for each team they were on. This dataset had 180 features before feature selection.

IV. METHODOLOGY

To conduct our analysis and model development, we focused our efforts under the following conditions and focal points.

- Offensive Players only – the number of overall offensive statistics available in the data set was focused on offense-oriented players.
- Contract-year performance statistics will be key predictors for maximizing the value of the second contract – generally, players make additional effort to boost performance statistics in the final year of their contract, the contract-year, because their play and performance heavily influence the contract they will be offered.
- The response variable will be the total guaranteed amount for the contract.
- Rookie offensive players starting from 2011 will be the target records as they would be the first players directly affected by the landmark 2011 CBA changes.
- Models Developed: Linear Regression (for benchmarking), Random Forest, SVM, Gradient Boosting (XGB), and Generalized Boosting Regression (GBM) models, including Cross-Validation for performance evaluation.

Linear Regression: Fitting a line that best represents the relationship between contract guaranteed value (response) and performance predictor variables.

Random Forest: An ensemble of decision trees, which is a splitting of data based on conditions to predict contract value; potential for overfitting due to noisy data.

Support Vector Machine (SVM): Robust supervised machine learning algorithm used for classification and regression by maximizing the margin between boundary and nearest data points.

Gradient Boosting (XGB): Ensemble technique for sequentially building models, correcting errors from previous models to improve accuracy while reducing bias and variance but requires fine tuning.

Generalized Boosting Regression (GBM): Another ensemble method, that combines multiple, weaker models to create stronger ones. This method also seeks to increase accuracy over iterations by sequentially correcting errors on previous models.

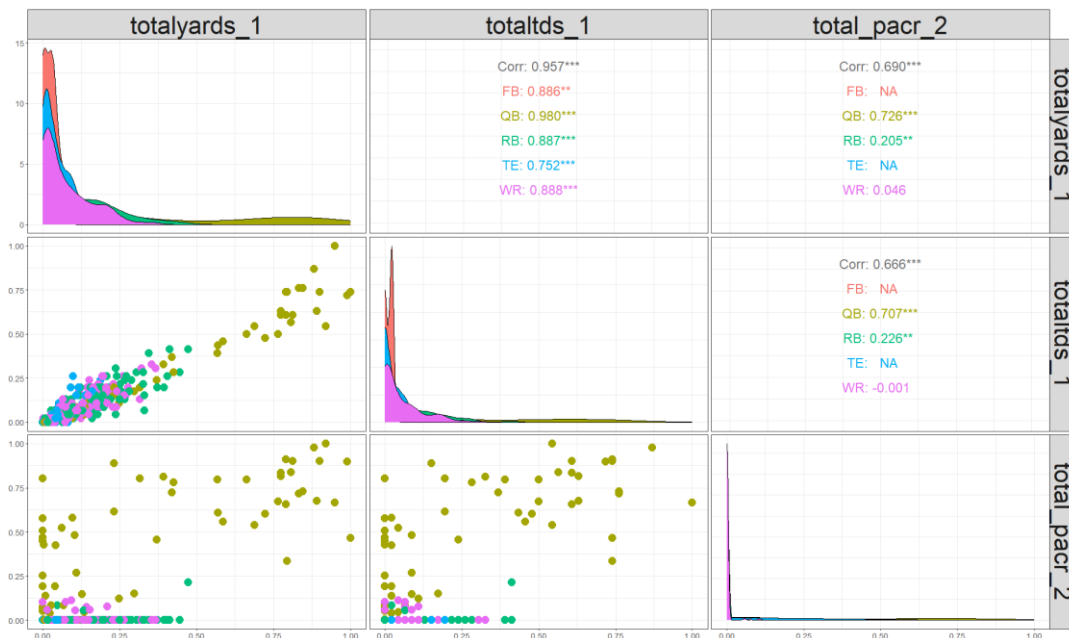
For hyperparameter tuning, 10-fold cross-validation was used with deterministic folds for each model in our analysis.

As a bonus, for our personal interest as football fans, we predicted the contract value for a set of players who have not yet signed a second contract. The results of our predictions can be found in the Model Validation section of the conclusion.

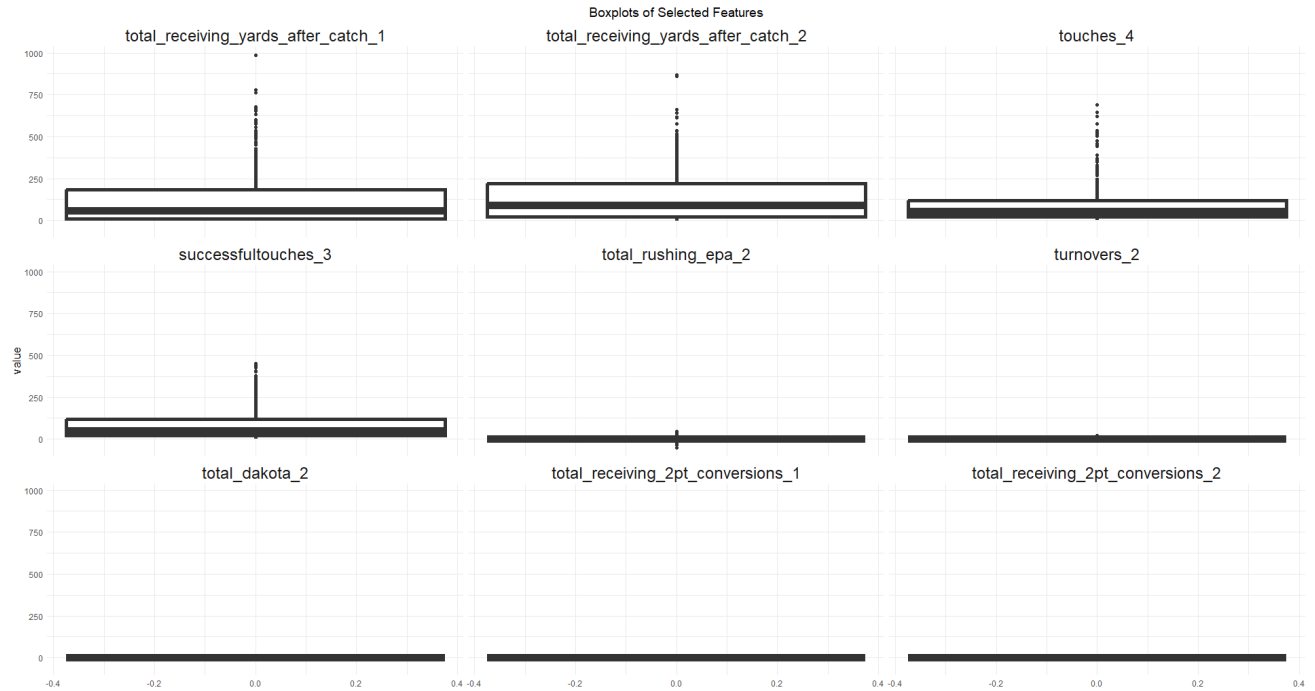
a. EXPLORATORY DATA ANALYSIS

Before modeling, it is imperative to gain a robust view of the feature set and the response. After widening the feature set as described in the Data Preparation section above so that each player has a single observation, the data set includes 646 observations, 180 features, and a single response *guaranteed*. Because of the size of the feature set, it will be necessary to reduce the size of the feature set to produce a model that balances simplicity with retaining information. Intuitively, there are many features that are obviously dependent. For example, `total_passing_yards_x` is intuitively dependent on

total_passing_air_yards_[x]. More formally, feature selection can begin by evaluating correlation plots to ascertain highly correlated features. Then, features that contain largely the same information as at least 1 other feature can be removed. The correlation matrix and pair plot below contain abbreviated versions of the collinearity evaluation that was performed to ultimately reduce the feature set. Then, for any pair of features where the correlation coefficient was at least 0.9, one of the features was eliminated. The pair plot displays not only the correlation coefficient between features, but the distribution of each and a scatter plot. From the pair plots shown below, totalyards_1 and totaltds_1 have a correlation coefficient = 0.967, have a linear relationship, and are heavily right skewed. In fact, one characteristic of much of the feature set is that they are heavily right-skewed, with many players having values close to 0 and a handful of large outliers for high-performing players.



To reiterate the skewed nature of much of the feature set, as well as the discrepancy in range, the boxplot below shows the distribution and range of values for selected features. A full boxplot of all features in the final data set after accounting for collinearity can be found in the supplemental files.



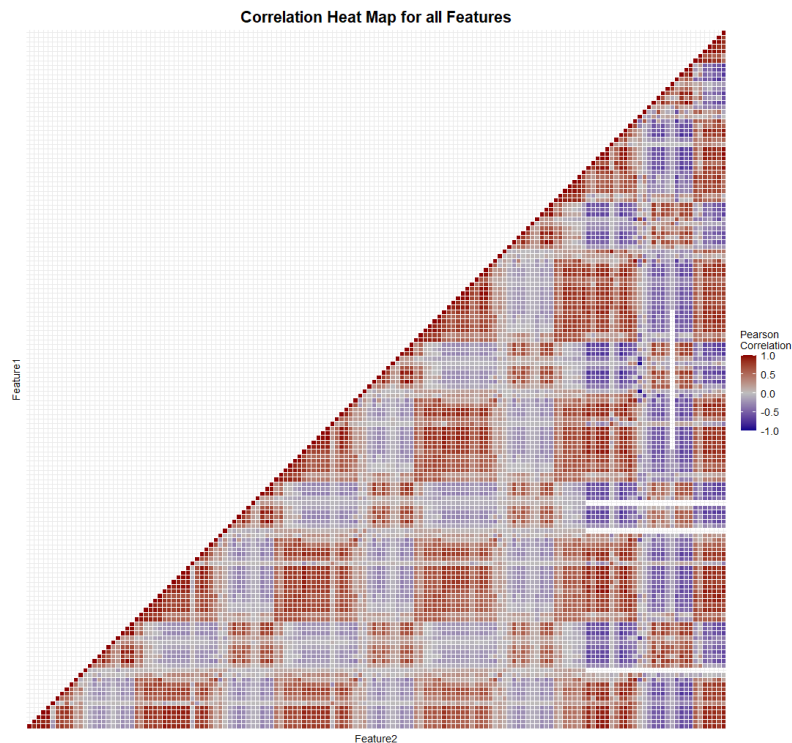
The boxplot above shows *total_receiving_yards_after_catch* and *touches* values that range anywhere from 0 to close to 1000, while features such as *total_receiving_2pt_conversions*, *total_dakota* and *turnovers* that are difficult to discern when presented on the same scale. In fact, *total_receiving_2pt_conversions* range from 0 to 2, *total_dakota* from -0.215 to 3.335 , and *turnovers* from 0 to 35. The table below additionally shows summary statistics for selected features to provide additional insight into the feature set and the response.

Summary Stats - Features (Abbreviated)

Statistic	total_receiving_yards_after_catch_4	total_receiving_2pt_conversions_4	total_racr_4	total_fantasy_points_ppr_4	touches_4	total_passing_epa_3	total_passing_2pt_conversions_3	total_pacr_3	total_rushing_fumbles_3	guaranteed
1 Min	0.000	0.000	-57.192	-1.640	1.000	-112.869	0.000	0.000	0.000	0.0000
2 Max	548.000	2.000	72.252	415.680	689.000	89.515	4.000	20.020	6.000	219.0100
3 Q1	18.750	0.000	0.802	29.325	22.000	0.000	0.000	0.000	0.000	0.0000
4 Median	70.000	0.000	6.325	66.900	53.000	0.000	0.000	0.000	0.000	0.2475
5 Mean	116.213	0.090	6.422	92.085	101.787	-0.906	0.081	1.164	0.524	10.2630
6 Q3	178.000	0.000	11.516	139.300	119.000	0.000	0.000	0.000	0.000	9.5750
7 Variance	14995.369	0.107	200.154	6385.150	17231.065	198.721	0.156	11.582	1.291	634.3990

From the table above, the discrepancy in range across features is apparent once again. Additionally, the difference between the mean and median of most features is material, providing further evidence to the skewed nature of the data. Then, observing the mean value across features, it is apparent that many features have largely small values; the mean of *total_receiving_2pt_conversions_4* and *total_passing_2_pt_conversions_3* is 0.090 and 0.081 respectively. Consequently, these features would have an immaterial impact on *guaranteed*.

Regarding collinearity, the correlation heat map displayed does not include the feature names for readability; however, it is still apparent that there are numerous pockets of highly correlated features. A correlation heat map with a subset of features and names displayed is included in the supplemental files.



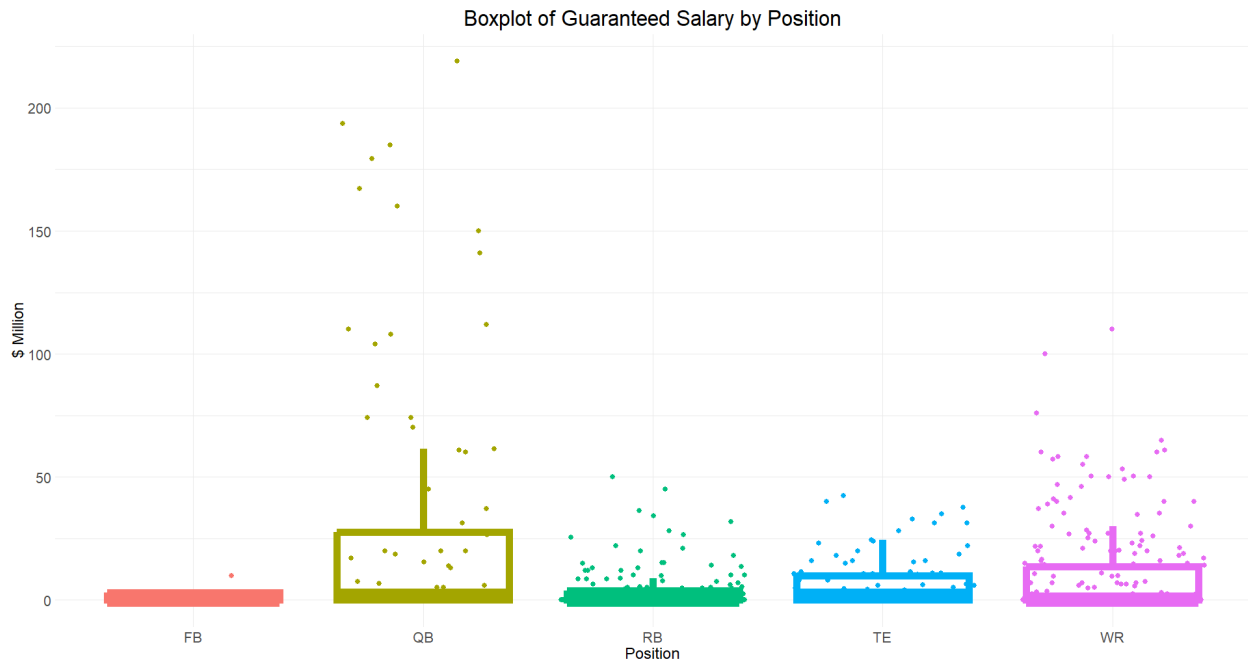
Another characteristic of the feature set is that there are many features that have many null values. In fact, there are only 2 observations with non-null values for the features ending in “_6”, which is applicable to features where the player had 6 years remaining on their contract. Also, there were 26 observations (4.0%) with data for 5 years remaining. Because of the relative non-existence of values for these features, ones ending in “_6” and “_5” were entirely eliminated from the feature set used in modeling. After accounting for collinearity and the features with mostly null values, only 45 predictors remain.

Null Observations by Years Remaining

	YearsRemaining	NA.Count
1	6	644
2	5	620
3	4	322
4	3	178
5	2	109
6	1	61

As for the response variable *guaranteed*, the distribution largely mirrors the feature set; the boxplot below shows a response that is right skewed with most values being around zero. Though the skewed

nature of the response exists across positions, it is much more pronounced for quarterbacks (QB). Intuitively, quarterbacks are largely considered not only the most valuable on-field position, but also the most known and marketable position off the field and, thus, warrant larger up-front investment.



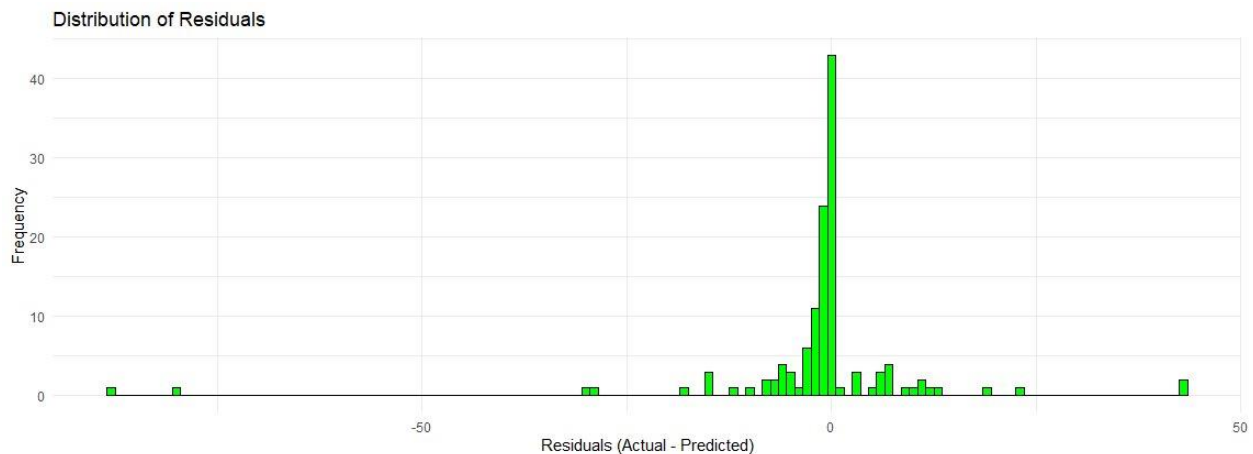
The boxplot also highlights the relative difference in range between quarterbacks and other positions. The extremely high values for guaranteed contracts for QBs could, therefore, be more difficult to predict. Because of the skewed nature of the feature set and the response, and because of the difference in range, it may be beneficial to normalize the data before modeling, particularly for models where the magnitude of values is impactful such as linear regression.

V. ANALYSIS AND RESULTS: LINEAR REGRESSION

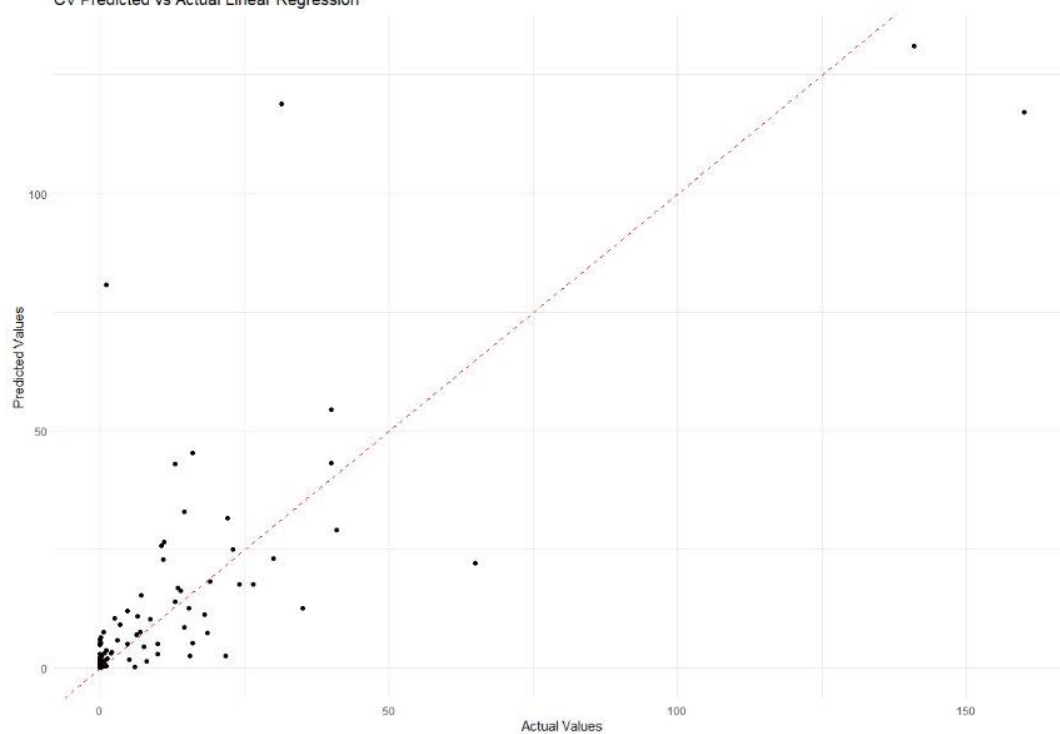
All the models below were tuned and their performance reported using the *caret* package in R. The linear regression model had root mean squared error (RMSE) of 16.107 and mean absolute error (MAE) of 8.822 on the test set. The R-squared and adjusted R-squared of the model were 0.797 and 0.770 respectively, while overall model p-value was $<2.2e-16$, indicating that the model does have explanatory power. At an alpha level of 0.001, the significant features were:

- *Total_passing_epa_2*
- *Total_passing_2pt_conversions_2*
- *Total_passing_epa_1*
- *Total_rushing_epa_1*
- *Total_receiving_epa_1*
- *Total_target_share_1*
- *Position.QB*

Since quarterbacks are generally viewed as the most important position and they typically receive the largest contracts, it's no surprise that being a quarterback is significant in the model. As seen by the distribution of residuals and the predicted versus actual values in the figures below, the linear regression model generally had accurate predictions but did not perform as well with larger contract values. This shows that there could be an issue of heteroskedasticity in the model as the residuals are not constant across the entire range of observations.



CV Predicted vs Actual Linear Regression

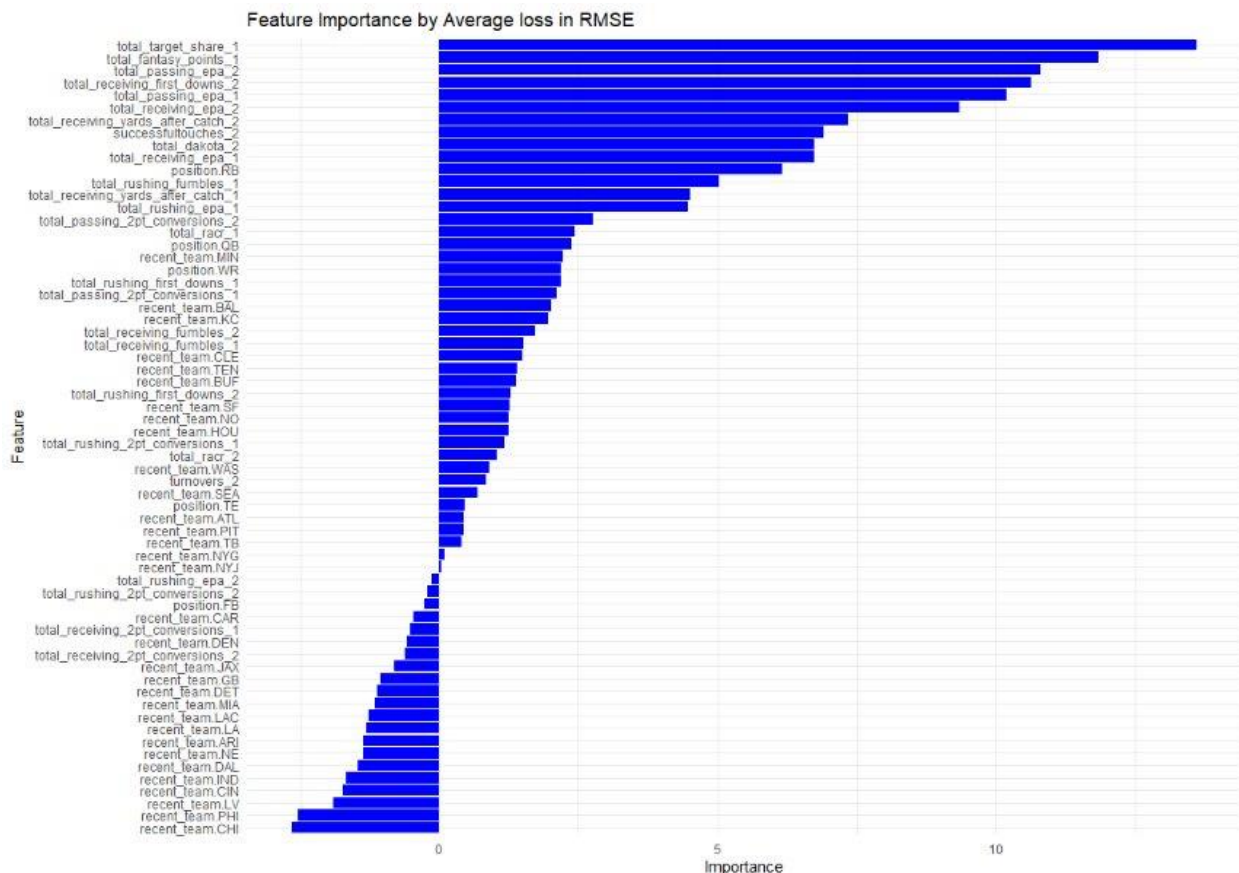


VI. ANALYSIS AND RESULTS: RANDOM FOREST

We use cross-validation to tune the hyperparameters of the random forest model, with the best model having the following values:

- `mtry = 5`
- Number of trees = 500

The random forest model using these parameters had the lowest RMSE and MAE of any of the models ran, with values of 13.481 and 5.530 on the test set respectively. As seen in the feature importance plot below, the most important feature in the random forest model was `total_target_share_1`, which is the number of passes that a player receives out of the team's total passes. Since this feature has the subscript 1, it is for the player's contract-year season. The distribution of residuals shows relative normality and the plot of predicted versus actual values shows accurate predictions, albeit with potential heteroskedasticity for larger contract values (distributions are included in the supplemental files).

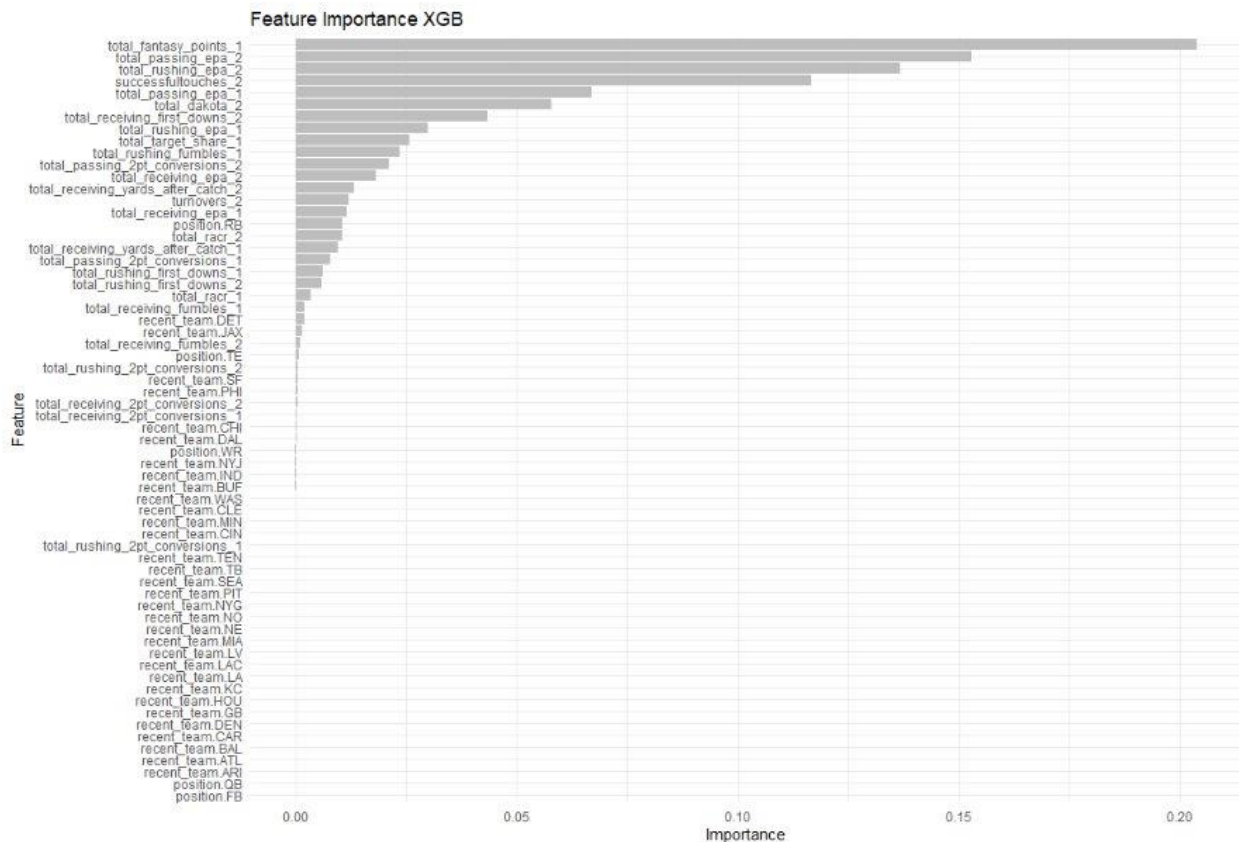


VII. ANALYSIS AND RESULTS: GRADIENT BOOSTING (XGB)

The optimal parameters from cross-validation for the gradient boosting were as follows:

- nrounds = 50
- max_depth = 4
- eta = 0.3

The gradient boosting model had the highest RMSE of all the models at 17.301 on the test set, along with an MAE of 6.306. The most important feature in this model was *total_fantasy_points_1*, which indicates a player's performance in the popular fantasy football game the season before they signed their second contract. The distribution of residuals shows relative normality and the plot of predicted versus actual values shows accurate predictions, albeit with potential heteroskedasticity for larger contract values (distributions are included in the supplemental files).

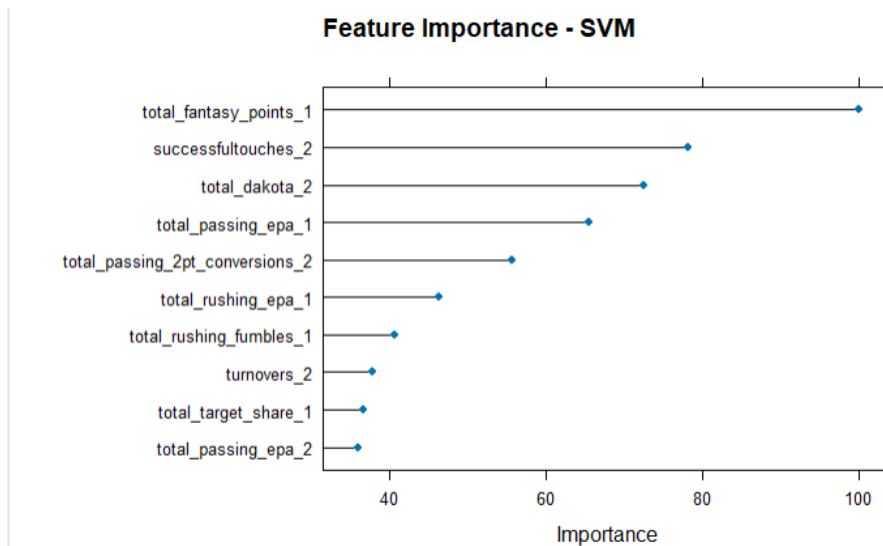


VIII. ANALYSIS AND RESULTS: SVM

Cross-validation produced the following optimal parameters for the SVM model:

- $\sigma = 0.01$
- $c = 10$

The SVM model had a moderate RMSE on the test set of 15.506, but an equal MAE value of 15.506 which was the worst MAE value of all the models. The most important feature in this model was *total_fantasy_points_1*. This was also the most important feature in the gradient boosting model, underscoring its importance in predicting the guaranteed value of a player's second contract. The distribution of residuals shows relative normality and the plot of predicted versus actual values shows accurate predictions, albeit with potential heteroskedasticity for larger contract values (distributions are included in the supplemental files).

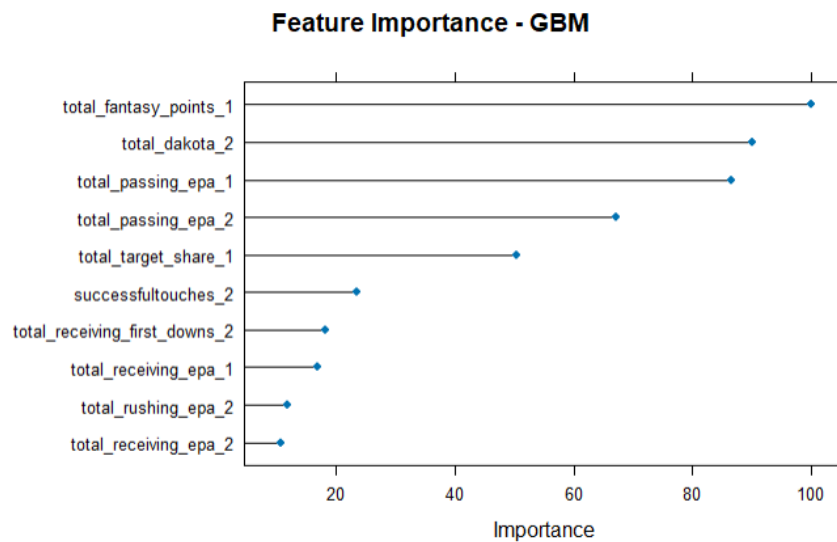


IX. ANALYSIS AND RESULTS: GBM

The GBM model had the following optimal parameters after cross-validation:

- trees = 50
- interaction.depth = 5
- shrinkage = 0.1
- minobsinnode = 10

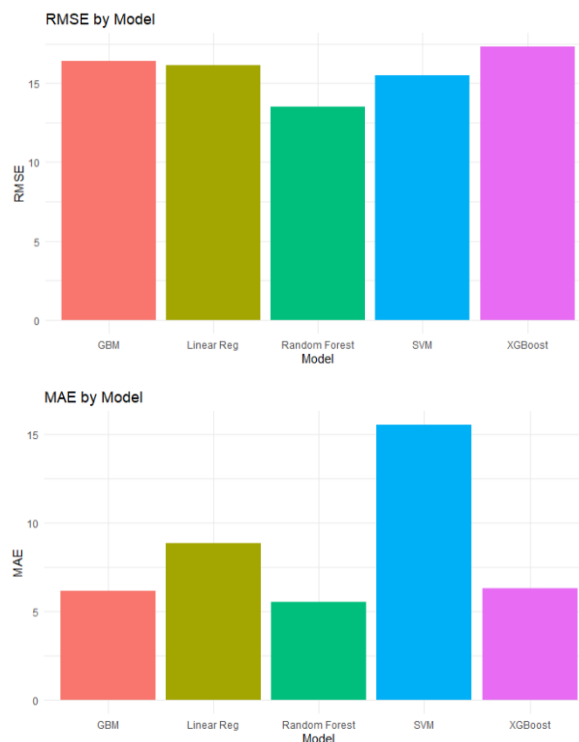
GBM performed moderately well with an RMSE of 16.410 and MAE of 6.157 on the test set. Similar to the gradient boosting and SVM models, the feature with the most importance was *total_fantasy_points_1*. The distribution of residuals shows relative normality and the plot of predicted versus actual values shows accurate predictions, albeit with potential heteroskedasticity for larger contract values (distributions are included in the supplemental files).



X. CONCLUSION

Our analysis of the guaranteed portion of an NFL player's second contract has provided valuable insights into the factors that influence NFL contracts. After examining numerous models such as linear regression, random forest, XGB boosting, SVM, and Generalized Boosting Regression (GBM), we identified key predictors to guaranteed value such EPA features and total fantasy points the year before signing a contract. EPA in the NFL refers to a player's expected points added, a metric used to quantify a player's individual impact on the team's ability to score. EPA benefits from weighing a player's performance in terms of team success, which is what team management ultimately values the most. Also, in the context of this analysis, EPA metrics are useful since they are measured for all positions, while some of the features included are generally specific to a given position (i.e. passing yards). Total fantasy football points are also advantageous because they are position-agnostic, at least for the offensive skill positions which were the only ones evaluated in this analysis. Lastly, features with the "1" suffix, indicating features measured for a player's final year of play before their succeeding contract, tended to have greater relative importance to each model than other seasons.

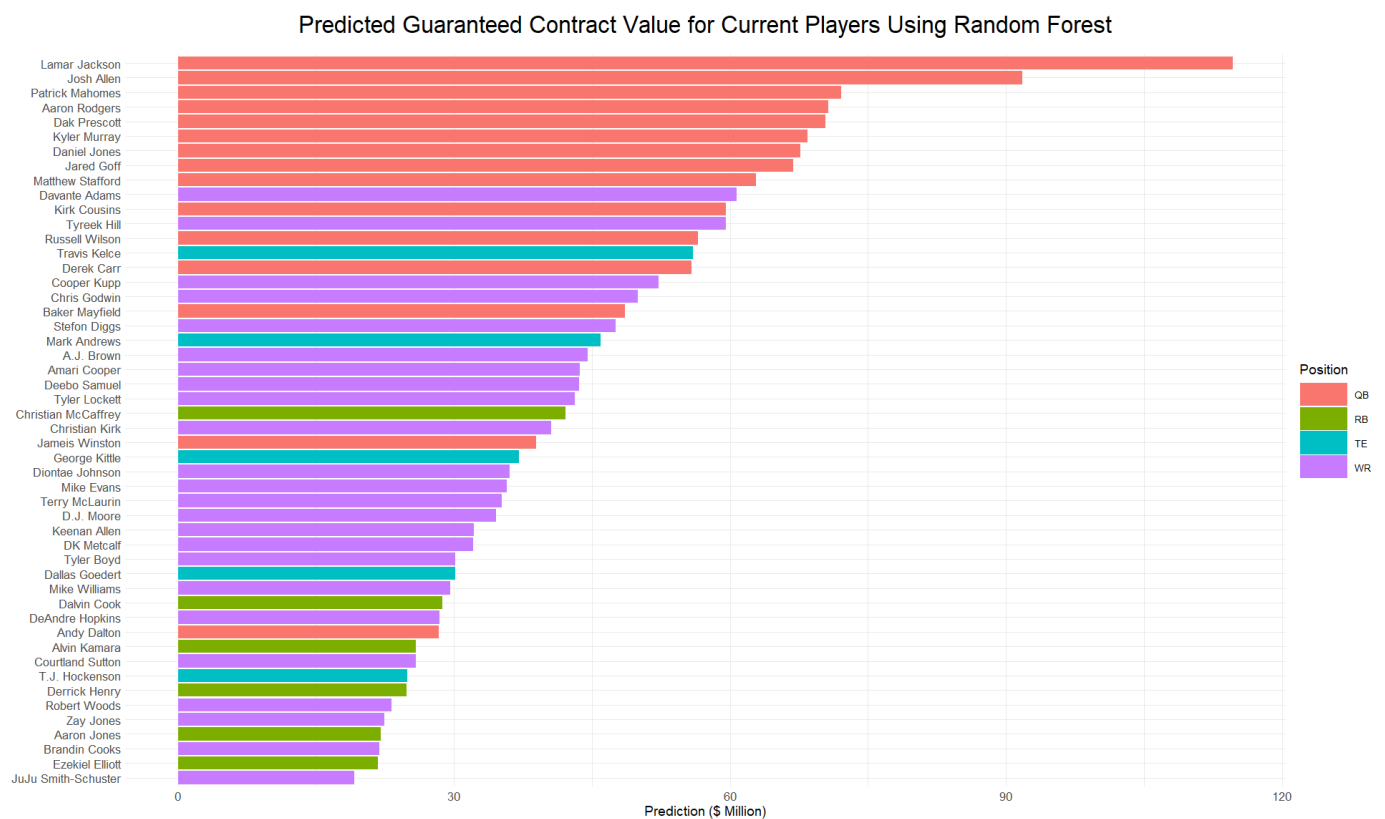
As displayed in the bar charts below, among the models examined, the random forest model demonstrated the best predictive performance in terms of both RMSE and MAE. Its ability to handle non-linear interactions between features and its robustness against outliers made it well-suited for the data set.



Our findings also reflect broader themes in modern sports: the critical role of data in decision-making, the value of advanced statistical metrics, and the importance of context in interpreting results. Future work could explore additional modeling techniques or incorporate more granular data, such as injury history or the player's height and weight to add additional information that could help improve predictability. Overall, this project has provided a robust, adaptable model to measure the guaranteed money a player should get to allow NFL teams to better allocate future resources efficiently and more effectively evaluate player performance in terms of their market value.

a. Model Validation

Any prediction model is more valuable when it can make accurate predictions on outside data sets. Because of the lack of available data beyond that used within this analysis, it is difficult to validate the accuracy of the optimal Random Forest model on an outside set. As a proxy, we used the Random Forest model with $mtry = 5$ to predict the guaranteed contract value for every active offensive skill position player (QB, RB, TE, WR, and FB). The validation assumes every active player is due to negotiate a new contract this season. The chart below shows the resulting top 50 predictions among active players.



Observing the results of the predictions for current players certainly adds confidence to the utility of the Random Forest model. Of the top 11 predicted values, 10 are quarterbacks, which is an intuitive result. Additionally, the drop-off from the top few players to the rest of the league mirrors the skewed nature of the training set. Lastly, as most NFL fans would agree, the names on the top 50 predicted guaranteed contracts are those who are known to be the top performing players.

XI. APPENDIX

Supplemental File Index

File Name	Description
Boxplot all features.png	Exploratory data analysis - boxplots for all features
Correlation heat map (selected).png	Exploratory data analysis – correlation heat map of all features
Data Dictionary – Contracts nflreadr.pdf	Data dictionary for Contracts repository
Data Dictionary – Draft Picks nflreadr.pdf	Data dictionary for Draft Picks repository
Data Dictionary – Player Stats nflreadr.pdf	Data dictionary for Player Stats repository
Data Preparation.R	R code for initial data wrangling and filtering
Group 44 Project.R	R code for data extraction, cleaning, combining, EDA, and valuation; used after the data preparation code
R Notebook.pdf	R code for model development and analysis
Summary Stats – Features (Full).1.png	Summary Stats of Features 1 of 3
Summary Stats – Features (Full).2.png	Summary Stats of Features 2 of 3
Summary Stats – Features (Full).3.png	Summary Stats of Features 3 of 3
Actual vs Predicted- Gradient Boosting.png	Scatter plot showing actual vs predicted values from gradient boosting model
Actual vs Predicted- Random Forest.png	Scatter plot showing actual vs predicted values from random forest model
Actual vs Predicted- GBM.png	Scatter plot showing actual vs predicted values from GBM model
Actual vs Predicted- SVM.png	Scatter plot showing actual vs predicted values from SVM model
Residual Plot – Gradient Boosting.PNG	Histogram showing residuals in gradient boosting model
Residual Plot – GBM.PNG	Histogram showing residuals in GBM model
Residual Plot – Random Forest.PNG	Histogram showing residuals in random forest model
Residual Plot – SVM.PNG	Histogram showing residuals in SVM model

XII. BIBLIOGRAPHY AND CREDITS

NFLverse

Carl S, Baldwin B, Sharpe L, Ho T, Edwards J (2023). *nflverse: Easily Install and Load the 'nflverse'*.
<https://nflverse.nflverse.com/>, <https://github.com/nflverse/nflverse>.